

The M_{24} Permutation Sigma-MGF

Exact class formula, orbit proof, and the degree-six boundary

Computational verification note

17 July 2026

Abstract

We prove the cycle-shape formula for the sigma-MGF of the natural 24-dimensional permutation representation of M_{24} . Two published ATLAS permutations are converted into 24×24 matrices. Independent generator-orbit enumeration is compared with a full conjugacy-class sum. Agreement holds in every tested case. The computation also sharpens the 5-transitivity statement: equality with S_{24} holds through total degree five, while $[m_{(6)}]$ is 12 rather than 11 and $[m_{(1^6)}]$ is 204 rather than 203.

1 Exact formula

Let g have cycle shape $\prod_{r \geq 1} r^{c_r(g)}$. On an r -cycle the eigenvalues are all r th roots of unity, so

$$\det(I - tP_r) = \prod_{j=0}^{r-1} (1 - t\zeta_r^j) = 1 - t^r.$$

Multiplying over cycles and averaging the generalized Molien integrand gives

$$\boxed{\mathcal{S}_{M_{24}, \text{perm}}(\mathbf{t}) = \sum_C \frac{|C|}{|M_{24}|} \prod_{i \geq 1} \prod_{r \geq 1} (1 - t_i^r)^{-c_r(C)}}. \quad (1)$$

This is an all-degree identity, not an asymptotic expansion.

The coefficient of m_τ is

$$\dim \left(\bigotimes_j \text{Sym}^{\tau_j} \mathbb{C}^{24} \right)^{M_{24}}.$$

Each symmetric power has a basis indexed by multisets. A permutation-group fixed vector is constant on each orbit of the resulting basis. Therefore

$$[m_\tau] \mathcal{S}_{M_{24}, \text{perm}} = \# \left(M_{24} \setminus \prod_j \text{MSet}_{\tau_j}(\{1, \dots, 24\}) \right). \quad (2)$$

Equations (1) and (2) are mathematically equivalent, but they produce genuinely different algorithms: weighted class expansion versus graph traversal under generators. Their agreement is the central cross-check.

2 Why equality with S_{24} stops after degree five

An S_{24} orbit of a product of multisets is determined by the multiplicity profile carried by its support points. If $|\tau| \leq 5$, that support has at most five points. The 5-transitivity of M_{24} maps any ordered listing of one support to the corresponding listing of another, so every S_{24} orbit remains a single M_{24} orbit. Consequently

$$\mathcal{S}_{M_{24}, \text{perm}} \equiv \mathcal{S}_{S_{24}, \text{perm}} \pmod{\deg > 5}.$$

For one symmetric power, the S_{24} coefficient is the partition number $p(k)$; for (1^k) it is the Bell number B_k (provided $k \leq 24$). The claim is not extended above degree five. Indeed, generator traversal finds two orbits on six-element subsets. This splits the all-distinct equality pattern and creates the first extra orbit.

3 Constructed matrices and checks

The verifier uses the ATLAS standard generators

$$\begin{aligned} a &= (1, 4)(2, 7)(3, 17)(5, 13)(6, 9)(8, 15)(10, 19)(11, 18) \\ &\quad (12, 21)(14, 16)(20, 24)(22, 23), \\ b &= (1, 4, 6)(2, 21, 14)(3, 9, 15)(5, 18, 10)(13, 17, 16)(19, 24, 23). \end{aligned}$$

They are converted to permutation matrices by $P_g e_j = e_{g(j)}$. The code asserts the standard-generator fingerprints

$$o(a) = 2, \quad o(b) = 3, \quad o(ab) = 23, \quad o(abababbababbabb) = 4,$$

and checks that the aggregated class sizes sum to $|M_{24}| = 244,823,040$.

For a class shape, the coefficient of t^k in $\prod_r (1 - t^r)^{-c_r}$ is obtained with exact integer convolution. In parallel, breadth-first traversal acts with $a^{\pm 1}, b^{\pm 1}$ on every multiset through degree six and every ordered tuple through degree four. At degree six the tensor calculation uses equality patterns together with the directly enumerated two orbits of six-element subsets.

coefficient	direct orbits	class formula	S_{24}
$m_{(1)}$	1	1	1
$m_{(2)}$	2	2	2
$m_{(3)}$	3	3	3
$m_{(4)}$	5	5	5
$m_{(5)}$	7	7	7
$m_{(6)}$	12	12	11
$m_{(1^2)}$	2	2	2
$m_{(1^3)}$	5	5	5
$m_{(1^4)}$	15	15	15
$m_{(1^6)}$	204	204	203

The degree-one invariant is the all-ones line. The degree-six discrepancies are a useful regression test: a program that silently substitutes 5-transitivity beyond its valid range returns the wrong values 11 and 203.

4 Reproduction and provenance

Run

```
.venv/bin/python for_this_guy/sporadic_matrix_groups/
  m24_permutation/verification.py
.venv/bin/marimo edit for_this_guy/sporadic_matrix_groups/
  m24_permutation/notebook.py
```

after joining wrapped lines. The full cycle-shape/class-size table is stored as exact integers in `sporadic_matrix_groups/common.py`; classes with the same permutation cycle shape are aggregated because (1) cannot distinguish them.

class	aggregated size	permutation cycle shape
1A	1	1 ²⁴
2A	11,385	1 ⁸ 2 ⁸
2B	31,878	2 ¹²
3A	226,688	1 ⁶ 3 ⁶
3B	485,760	3 ⁸
4A	637,560	2 ⁴ 4 ⁴
4B	1,912,680	1 ⁴ 2 ² 4 ⁴
4C	2,550,240	4 ⁶
5A	4,080,384	1 ⁴ 5 ⁴
6A	10,200,960	1 ² 2 ² 3 ² 6 ²
6B	10,200,960	6 ⁴
7AB	11,658,240	1 ³ 7 ³
8A	15,301,440	1 ² 248 ²
10A	12,241,152	2 ² 10 ²
11A	22,256,640	1 ² 11 ²
12A	20,401,920	24612
12B	20,401,920	12 ²
14AB	34,974,720	12714
15AB	32,643,072	13515
21AB	23,316,480	321
23AB	21,288,960	123

The sizes sum to 244,823,040. Source: Online ATLAS of Finite Group Representations, retrieved 18 July 2026. The SHA-256 of the canonical serialized table is

103ba576555f01bea74df73f8e2f788b02bee4e8b20df3ed198eb005d16de044.

The SHA-256 of the software data file is

3fbc8e5cde4ac5670696ecc1c5f692a2afb2065469a83ce486aa8b28d046d579.

Generator-orbit enumeration is independent of the class table. The class-sum calculation is an exact cross-check conditional on the externally supplied ATLAS data; the complete table and checksums make that input auditable from this PDF.

Generator data and the group order are from the online ATLAS of Finite Group Representations, <https://brauer.maths.qmul.ac.uk/Atlas/spor/M24/>. The calculation verifies the stated representation and coefficients without enumerating all 244 million group elements.